

Intro to Cloud Computing (CS4037) BCS9A, BDS7A

Date: September 24th, 2024

Course Instructor(s): Ali Khawaja

Sessional-I Exam

Total Time (Hrs.): 1

Total Marks: 100

Total Questions: 9

Roll No

Section

Student Signature

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Attempt all the questions. Provide concise and brief answers.

CLO 1. Apply cloud computing knowledge in various real world use cases (situations)

Q1: Briefly explain how an organization can connect its on-premises network and its Azure Network. Briefly, and accurately provide steps required to perform this type of configuration. (10)

CLO 2. Apply cloud computing knowledge in various real world use cases (situations)

Q2: When dealing with containers, clearly and briefly explain the usage and benefits of Azure Webapp, ACI, and AKS. (20)

CLO 1. Describe the basics of cloud computing and related services

Q3: Explain the concepts of fault domains and update domains and what specific problems they solve in enterprise environments. Name the feature which allows you to setup and configure this functionality? (10)

CLO 2. Apply cloud computing knowledge in various real world use cases (situations)

Q4: Explain briefly what problem Virtual Machine scale sets solve? Can we have same or different types of VMs in VM Scale sets? How VMSS ensures that new VMs will be created with the required software and configuration? (10 Points)

CLO 1. Describe the basics of cloud computing and related services

Q5: Provide answers for the below (10 Points)

1. An Azure service-level agreement (SLA) guarantees which of the following metrics? (2.5 Points)
 - a. Security
 - b. Resiliency
 - c. Performance
 - d. Uptime
2. Organization ABC is based in the United States and wants to take advantage of cloud services and Hyperscale. However, Organization ABC must abide by strict government requirements. ABC Organization plans to adopt Azure Private Cloud and use it internally and to consume public cloud

services using the Azure cloud in the United States. Provide the name of the cloud deployment model ABC Organization should adopt.

3. What is the difference between regular windows 10/11 and multi-sessions windows 10/11?
4. Which Azure resource holds related resources in an organization's Azure cloud subscription?

CLO 2. Apply cloud computing knowledge in various real world use cases (situations)

Q6: What are A and CNAME records in DNS? Provide an example for each record type. Briefly describe the steps required to do private/internal DNS resolution in Azure VNETs. (10)

CLO 1. Describe the basics of cloud computing and related services

Q7: Specify whether each of the below statements are true or false (10 Points)

1. The shared responsibility model is heavily tied into the cloud service types.
2. The general public availability is a key difference between public and private clouds.
3. Cloud computing falls under OpEx because cloud computing operates on a consumption-based model.
4. Cloud computing is a way to rent compute power and storage from someone else's datacenter.
5. Multi-cloud model uses some datacenters focused on providing cloud services to anyone that wants them, and some data centers that are focused on a single customer

CLO 2. Apply cloud computing knowledge in various real world use cases (situations)

Q8: Explain the SAAS and IAAS parts of AVD. How does AVD provides enhanced security. (10)

CLO 2. Apply cloud computing knowledge in various real world use cases (situations)

Q9: Divide VNET address space of 192.168.0.0/22 into four equal address spaces. (10)

BONUS Question

CLO 2. Apply cloud computing knowledge in various real world use cases (situations)

Q10: Your organization has one VNET in UAE North region and one VNET is Central USA region. Provide the most optimal way to connect both VNETs with steps to perform that configuration. (10)